

MarketMind AI — Production Readiness Checklist

What's missing to take this from working demo to real-world use. List only — nothing here has been implemented yet.

Security — the biggest gap

- No auth anywhere. Any URL that can reach the FastAPI/Streamlit app can view all customer PII, trigger real emails, retrain/promote models. Needs a login layer at minimum.
- No rate limiting — nothing stops someone spam-clicking "send campaign" or "trigger retrain."
- Secrets live in a local .env; production needs a real secrets manager, not a file.
- DB password is a placeholder default; must be a real credential outside localhost.

Email / compliance

- Mailgun sandbox domain: caps at 5 authorized recipients, 300 emails/day, "sandbox" in headers. Real sending needs a verified custom domain (SPF/DKIM/DMARC DNS records).
- No unsubscribe link/mechanism — legally required (CAN-SPAM/GDPR) for marketing email.
- No bounce/complaint/spam-report handling, and campaign_log.opened/.clicked need the Mailgun webhook wired to actually populate.

Model

- The current Production champion (v1) is the leaky, inflated one (AUC 0.999) — should be replaced with an honestly-trained challenger before this drives real decisions.
- No scheduled retraining (manual/button-triggered only) and no live monitoring of prediction accuracy as real churn outcomes come in — only the training-time PSI check.
- Feature engineering isn't true point-in-time-correct (documented, known limitation).

Data

- No real ingestion pipeline — only loads from local CSVs. A live CRM/e-commerce data source integration doesn't exist.
- No data retention/consent policy for real customer PII.

Infra / ops

- MLflow on local SQLite — not multi-user-safe, no backups. Needs a real tracking server + DB + artifact store for production.
- No DB migrations (schema changes are ad hoc create_all, no Alembic).

- No logging/error-tracking/metrics stack (structured logs, Sentry-style error tracking, latency dashboards).
- github_actions.yml only runs tests — no actual build/deploy pipeline.
- Docker images aren't pushed anywhere; no real deploy target confirmed working end-to-end (Render config is untested against a live account).
- Streamlit isn't built for high-concurrency multi-tenant traffic — fine as an internal tool, a real constraint at scale.

Business logic

- Compliance checks are a stub (discount cap + keyword regex) — needs real legal review.
 - No feedback loop closing "did this campaign actually reduce churn" back into the model.
 - Product recommendations are popularity-based, not personalized.
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